POWER PULSE

FACTS Important Formulae

Notes by Balaji Sir

1 Series compensation

1. $$k = \frac{X_c}{X_L}, \quad X_{eff} = X_L(1-k), \quad P_{max} = \frac{V_1 V_2}{X_L(1-k)}$$

2. $$f_{er} = f_0\sqrt{\frac{X_c}{X_L}}, \qquad f_{mech} = f_0 - f_{er} \ \text{(SSR)}$$

sub-synchronous resonance

3. $$X_{TCSC} = -\frac{X_c X_L(\alpha)}{X_L(\alpha) - X_c}, \qquad K_{boost} = \frac{X_L(\alpha)}{X_L(\alpha) - X_c}$$

2 Shunt compensation

4. $$Q_{SVC} = V^2 B, \qquad \text{TCR: } B_{TCR} = \frac{\sigma - \sin\sigma}{\pi X_L}, \ \ \sigma = 2(\pi - \alpha)$$

firing angle α: 90° (full) to 180° (off)

5. $$\text{SVC: } Q \propto V^2; \qquad \text{STATCOM: } Q = V\, I_{max} \ (\propto V)$$

STATCOM supports the voltage better during dips

6. $$P = \frac{2V^2}{X}\sin\frac{\delta}{2} \ \text{(ideal mid-point shunt compensator)}$$

P_max = 2V²/X

7. $$V = V_{ref} + X_{sl}\, I_{SVC}$$

SVC voltage-current characteristic with slope X_sl

3 Phase shifter and UPFC

8. $$P = \frac{V^2}{X}\sin(\delta \pm \varphi)$$

phase-angle regulator controls real power

9. UPFC = STATCOM + SSSC on a common DC link: controls P, Q and V

10. SSSC: injects a series voltage in quadrature with the line current

11. Harmonics: delta-connected TCR removes triplen harmonics; 5th, 7th, 11th, 13th remain